

A Survey of Knowledge Graph Enhanced Retrieval-Augmented Generation

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Abstract—Factual reliability is a critical requirement in Education Technology (EdTech), especially as Large Language Models (LLMs) are increasingly used for personalized tutors. LLMs alone often suffer from factual inaccuracies (hallucinations) and static, outdated knowledge. While Retrieval-Augmented Generation (RAG) was developed to ground LLMs in external data, standard RAG implementations using unstructured text still struggle to provide complete context for complex, domain-specific questions. This limitation hinders their use in high-stakes fields. This paper surveys an emerging solution: Knowledge Graph Enhanced RAG (KG-RAG). We examine and categorize contemporary KG-RAG frameworks from fields like education, cybersecurity, and biomedicine, organizing them by architectural strategy (e.g., hybrid retrieval, graph based context expansion). Our findings show that KG-RAG systems consistently surpass baseline RAG performance, citing quantitative evidence such as a 35.2% increase in student assessment scores and an 18.7% accuracy improvement in complex biomedical Question and Answering (QA). This synthesis of current methods and open challenges aims to guide researchers in developing more dependable, domain-aware Artificial Intelligence (AI) systems.

Index Terms—Retrieval-Augmented Generation, Knowledge Graphs, Large Language Models, Domain Specific Question Answering, Factual Consistency

I. INTRODUCTION

Innovation in Education Technology (EdTech) is increasingly propelled by the integration of digital platforms with Artificial Intelligence (AI) [1] and Natural Language Processing (NLP) [2]. Large Language Models (LLMs) [3], [4] are central to this shift, demonstrating a powerful capacity for human-like text comprehension and generation. This capability is fueling new tools, such as adaptive tutors and automated question-generation systems, with applications also appearing in other data-heavy domains [5], including cybersecurity [6]. The EdTech sector's goal of providing flexible and personalized learning [5] is complicated by the rapid expansion of diverse educational materials, which poses serious challenges for maintaining relevance and pedagogical coherence [6].

The fluency of LLMs belies significant limitations that hinder their application in critical domains. Their knowledge

is static, fixed at the point of training, which can lead to outdated responses. A more serious issue is their tendency for "hallucination" [7], where they produce factually incorrect yet plausible-sounding statements. Furthermore, their internal reasoning is often opaque [8], [9]. Retrieval-Augmented Generation (RAG) was introduced to counter these issues by connecting the LLM [10] to external, verifiable information [9], [11]. Yet, standard RAG approaches, which generally use vector search on raw text, present their own difficulties [12]. Such methods can retrieve shallow or fragmented context, do not effectively synthesize information from multiple sources, and perform poorly on queries needing complex, multi-step reasoning [8], [11], [13].

Knowledge Graphs (KGs) [14] present a potential solution to these RAG limitations. A KG models information as a structured network of entities and their defined relationships, offering an explicit map of knowledge [15]. By incorporating KGs into the RAG process (forming KG-RAG systems), it is possible to improve retrieval context, support the multi-hop reasoning [6] that is difficult for standard RAG, and generate answers that are both more accurate and interpretable [16]. This quickly developing area requires a systematic review to organize current findings and direct future work.

This paper delivers a wide-ranging survey of the KG-enhanced RAG field. Its main contributions include a detailed review of core concepts and significant KG-RAG models from various domains, complete with summaries of their architecture, implementation, and empirical results. We also present a critical analysis of the comparative advantages and disadvantages of current methods, including hybrid retrieval, graph-guided context generation, and novel prompting strategies. This examination confirms the effectiveness of the KG-RAG approach, which has demonstrated consistent and substantial gains over baseline models. Specific examples include a 35.2% improvement in student assessment scores [5] and an 18.7% increase in biomedical QA accuracy [16]. The paper concludes by synthesizing these results to identify key unresolved challenges and propose avenues for future work on

domain-specific, dependable AI systems.

To ensure a comprehensive and contemporary review, we adopted a structured methodology for paper selection. We primarily queried academic databases including IEEE Xplore, ACM Digital Library, and the arXiv preprint repository for publications spanning from 2023 to early 2025, reflecting the newness of the field. Key search phrases included combinations of "Retrieval-Augmented Generation," "Knowledge Graph," "KG-RAG," and "Graph-based RAG." Our inclusion criteria prioritized peer-reviewed works and significant preprints that detail novel architectures, present empirical results, or apply KG-RAG to specific domains like biomedicine and cybersecurity. This process yielded the set of core frameworks analyzed in this survey.

The remainder of this paper is organized as follows. Section II provides a background study, detailing the architectures, implementations, and novel contributions of state of the art KG RAG frameworks, and concludes with a comparative summary. Section III analyzes the experimental results and performance improvements reported by these frameworks. Finally, Section IV concludes the survey by synthesizing key findings and highlighting promising directions for future research.

II. BACKGROUND STUDY

Creating dependable, domain-specific AI is essential for fields such as EdTech [5] and cybersecurity [6]. The research trajectory has moved from initial LLMs, which are prone to hallucination [7] and opaque decision making [8], [9], toward standard RAG [10]. This conventional RAG, however, often fails to deliver adequate context for intricate, multi-step queries when using unstructured text sources [8], [11], [13]. This specific inadequacy has spurred the creation of KG-enhanced RAG, which uses structured KGs to improve contextual understanding and reasoning [6], [14], [16]. Our survey aims to provide a structured summary of this area by classifying the architectures of prominent KG-RAG systems.

Figure 1 presents a taxonomy of the KG-RAG systems we reviewed, classifying them by their fundamental architectural design. This categorization illustrates the varied methods being used to combine structured knowledge with LLMs, with each design choice reflecting a different compromise within the RAG pipeline. These methods vary: some are hybrid retrieval models blending the flexibility of vector search with the accuracy of graph queries (e.g., Failure Mode and Effects Analysis (FMEA KG-RAG), Cybersecurity Knowledge Graph-Enhanced Retrieval-Augmented Generation (CyKG-RAG)), while others utilize the graph as the sole, precise retrieval source (e.g., Knowledge Graph based Retrieval-Augmented Generation (KGRA)). Alternative strategies involve using the graph to expand context for richer prompts (e.g., KG2RAG), to steer complex reasoning (e.g., Knowledge Graph Enhanced RAG framework (KRAGEN)), or to support specialized generation tasks (e.g., KAQG). This classification offers a clear framework for analyzing the field's developing methods.

A. Key Methodologies and Formulas

Several mathematical rules support the techniques used in RAG and KG-RAG systems. Sparse retrieval methods often rely on term-frequency algorithms like BM25, while dense retrieval is based on vector similarity metrics.

The BM25 scoring function [9] is a standard for sparse retrieval, calculating the relevance of a document (or chunk C_i) to a given query, as shown in equation (1).

$$\text{Score}(q, C_i) = \sum_{q_j \in q} \text{IDF}(q_j) \cdot \frac{f(q_j, C_i) \cdot (k_1 + 1)}{f(q_j, C_i) + k_1 \cdot (1 - b)} + k_1 \cdot b \cdot \frac{|C_i|}{\text{avg_dl}} \quad (1)$$

where IDF is the inverse document frequency, $f(q_j, C_i)$ is the term frequency, and k_1 and b are hyperparameters.

For dense retrieval, Cosine Similarity [14] is a widely used metric to measure the similarity between the vector representation of a query (v_q) and a chunk (v_c), formulated in equation (2).

$$\text{Similarity}(v_q, v_c) = \cos(\theta) = \frac{v_q \cdot v_c}{\|v_q\| \|v_c\|} \quad (2)$$

Within KG-centric applications, such as the KAQG framework for education, algorithms like PageRank [15] are used to determine the importance of concepts (nodes) within the graph. The formula is given in equation (3).

$$PR(v) = (1 - d) + d \sum_{u \in In(v)} \frac{PR(u)}{\text{Out}(u)} \quad (3)$$

where $In(v)$ are the inbound links to node v , and d is a damping factor.

Furthermore, in specialized educational applications, Item Response Theory (IRT) [15] is used to model the probability of a correct response based on item difficulty (b), discrimination (a), and guessing (c) parameters, as defined in equation (4).

$$P(\text{correct}) = c + \frac{1 - c}{1 + e^{-a(\theta - b)}} \quad (4)$$

B. Frameworks and Architectures

The following frameworks highlight diverse approaches to integrating KGs into RAG pipelines across various domains.

The **WeKnow-RAG** framework [9] employs a two-part pipeline that combines multi-stage web retrieval with structured KG queries (see Fig. 2). Its most novel component is a self-assessment function, allowing the LLM to gauge its own confidence level to minimize hallucinations. This feature was instrumental in its 3rd-place finish at the Meta KDD CUP 2024.

The self-assessment module is a crucial innovation for production systems, as it directly addresses the problem of hallucination by quantifying uncertainty. This suggests a move towards more "self-aware" RAG pipelines.

For biomedical applications, **KRAGEN** [16] integrates a KG-supported vector database with **GoT** (Graph-of-Thoughts) prompting [17] (Fig. 3). The GoT method decomposes a problem into smaller parts and uses RAG to fetch facts for

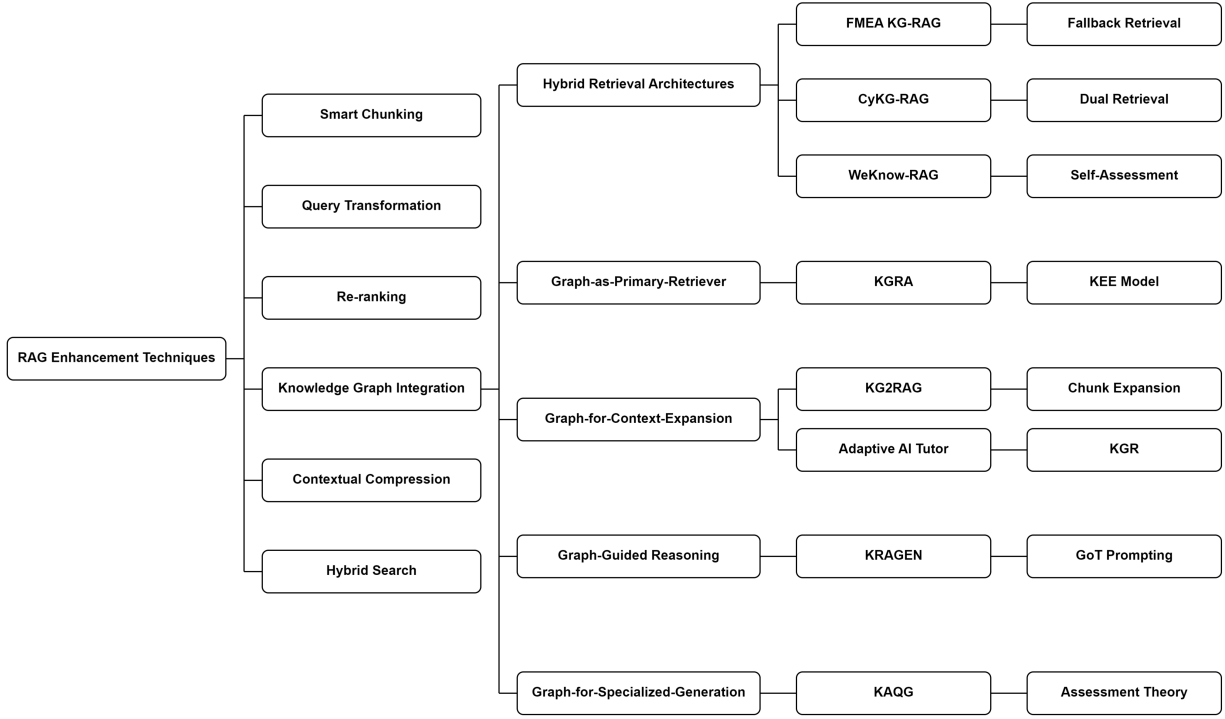


Fig. 1. A taxonomic overview of the KG-RAG techniques discussed in this survey.

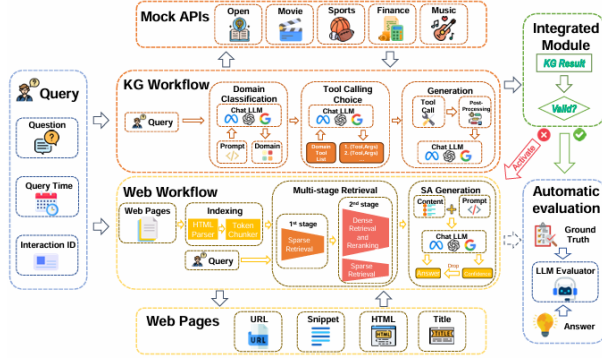


Fig. 2. The WeKnow-RAG pipeline, which combines a KG workflow with a web-based workflow and includes an automatic evaluation module (from [9]).

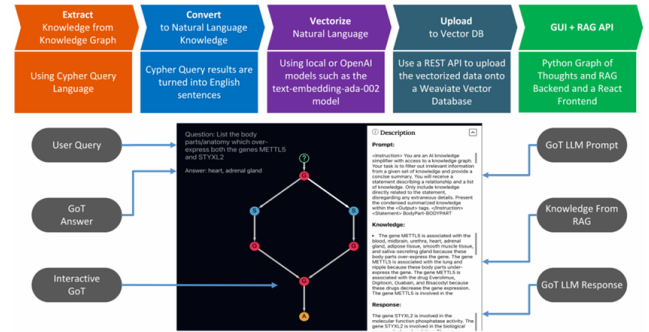


Fig. 3. Application flow of the KRAGEN framework, highlighting the use of Graph-of-Thoughts (GoT) prompting to guide the LLM’s reasoning process (from [16]).

each subproblem. This systematic reasoning process yielded substantial accuracy gains on multi-hop queries.

KRAGEN’s use of GoT demonstrates that the reasoning process (prompting) is as important as the retrieved data. The graph guides not just what to retrieve, but how to think.

The **KGRA** framework [8] is designed for domain-specific QA in situations lacking training data (“cold-start”) (Fig. 4). It features a novel, efficient, and precise **Key Entity Extraction (KEE)** [18] model built via data distillation. This KEE model translates user questions into exact Cypher queries, serving as an alternative to less precise vector search and reaching a 95% F1 score for keyword extraction.

This framework highlights a key trade-off: sacrificing the

"fuzziness" of vector search for the precision of graph queries. Its success in a cold-start setting (95% F1) is significant for domains where high-quality training data for vector models is unavailable.

In cybersecurity, **CyKG-RAG** [6] draws from two knowledge sources: a local, private KG (from logs) and a public, external CSKG (from CVE, MITRE, etc.). Its retrieval is hybrid, concurrently generating formal graph queries (SPARQL/Cypher) while also executing vector searches on KG node embeddings. This allows it to handle complex queries, such as those related to MITRE ATT&CK techniques.

The hybrid retrieval approach (graph queries + vector search) represents a practical compromise, blending the preci-

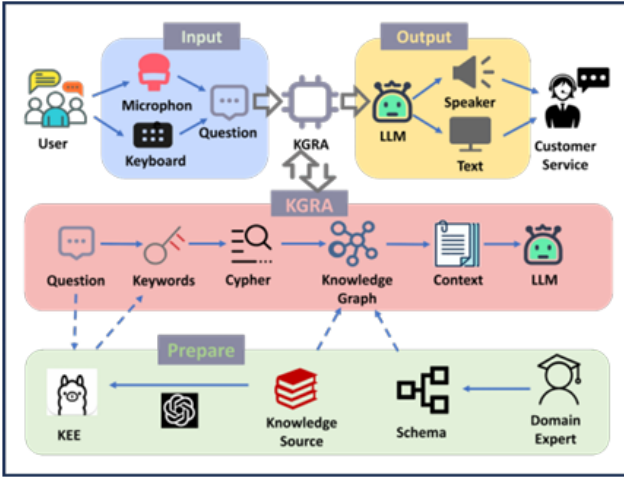


Fig. 4. The KGRA pipeline, which follows a "Question Keywords Cypher Context" flow to replace vector search with precise knowledge graph queries (from [8]).

sion of KGs with the broader coverage of vector embeddings. The use of two KGs (public/private) is a notable security architecture.

This framework for industrial FMEA [14] converts tabular FMEA data into a KG structure. It generates context-heavy chunks by performing a depth-first search (DFS) originating from 'FailureMode' nodes. The system first attempts retrieval with a direct Cypher query generated by an LLM, reverting to vector search only if the query fails. A usability study reported this method was substantially more practical than conventional tools.

This work proves KG-RAG's value in specialized industrial domains. The DFS-based chunking method traversing the graph to build context is more domain-aware than arbitrary fixed size text splitting.

The **KG2RAG** framework [11] is designed to enhance coherence in multi-hop QA. It begins with a standard semantic search to find "seed chunks." Then, a unique **KG-guided chunk expansion** process traverses the graph to locate related chunks, even if they are not semantically close. A concluding KG-based organization step removes noise, and this full approach consistently performed better than baselines on the HotpotQA dataset.

KG2RAG directly tackles the "fragmented context" problem of standard RAG. By using the graph to find related but not semantically similar chunks, it excels at multi-hop reasoning where vector search alone would fail.

An **Adaptive AI tutor** for EdTech [5] constructs a KG from course content (Fig. 5). Its inventive Knowledge Guided Retrieval (KGR) process identifies a pertinent node via semantic search and then traverses the graph to gather a well-connected, rich context, which overcomes the problem of fragmented information. A controlled experiment found that students who used this tutor achieved scores 35% higher ($p < 0.001$).

The 35% score increase provides strong quantitative evi-

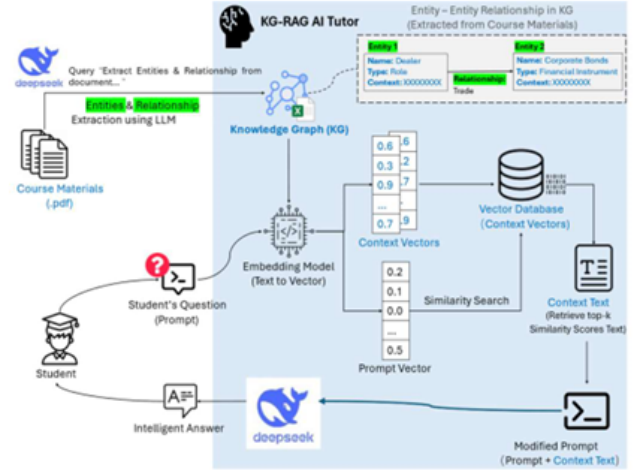


Fig. 5. The architecture of the KG-RAG AI Tutor, showing how a KG is used to build a KG which then enriches the retrieval process for student questions (from [5]).

dence for KG-RAG's real-world impact in EdTech. The KGR process, which gathers a connected subgraph, directly mirrors how an expert would gather context, rather than just isolated facts.

Also in the EdTech space, **KAQG** [15] is a multi-agent system that generates questions with controlled difficulty. It combines RAG with educational assessment principles. The system uses PageRank to identify key concepts and then creates questions aligned with Bloom's Taxonomy. The difficulty of these questions is subsequently validated with Item Response Theory (IRT), resulting in items that meet official ACT standards.

This framework moves beyond simple QA to a complex generation task (question generation). Its integration of PageRank (for importance) and IRT (for difficulty) shows a mature fusion of graph theory, NLP, and educational psychometrics.

The frameworks reviewed in this section present diverse strategies for KG-RAG implementation. Table I offers a detailed comparison and summarizes each framework's target domain, the datasets used, its method for graph construction, its specific retrieval strategy, the reported LLM backbone, and its most significant limitations or trade-offs. This analysis highlights the architectural choices made by researchers to balance precision, recall, and complexity.

III. EXPERIMENTAL RESULTS

Experimental evidence confirms that Knowledge Graph-enhanced RAG (KG-RAG) provides consistent performance benefits over traditional RAG and standalone LLM baselines. The observed improvements include better response accuracy, fewer hallucinations, and higher quality retrieved information. For instance, specific studies report that KG-RAG can increase student assessment scores by 35 percent, improve biomedical QA accuracy by over 18 percentage points, and reach far greater context precision (0.82 compared to 0.29). These ad-

TABLE I
COMPARATIVE ANALYSIS OF REVIEWED KG-RAG FRAMEWORKS (UPDATED FROM SOURCE PDFS)

Framework	Domain	Dataset(s)	KG Construction	Retrieval Strategy	LLM Backbone	Key Limitation
WeKnow-RAG [9]	General QA	CRAG Benchmark	Uses Mock KGs provided by benchmark via APIs	Hybrid: KG Tool-Calling + Multi-stage Web (Sparse + Dense)	Llama-3-70B-Instruct	Relies on external APIs and LLM's tool-use capability.
KRAGEN [16]	Biomedicine	AlzKB, BioASQ	Converts KG dump (Neo4j) to vectors in a Weaviate DB	Graph-of-Thoughts (GoT) RAG on vector DB	Not Specified (Compares to BioGPT, OpenChat)	High complexity in GoT prompting; reasoning overhead.
KGRA [8]	Domain-Specific QA (Cold-start)	Custom (University Admissions)	Expert-defined ontology; GPT-4 extraction, manually refined	Graph-only: Custom KEE model generates Cypher query	glm4-9b (Generator) Llama-3-8B (KEE)	Replaces vector search; performance hinges on KEE model accuracy.
CyKG-RAG [6]	Cybersecurity	Public CSKGs (MITRE) & Local Logs	Rule-based (RML) for public data; LLM for local logs	Hybrid: LLM generates SPARQL/Cypher + Vector search	ChatGPT 3.5-Turbo	Complexity in fusing results from public & private KGs.
FMEA KG-RAG [14]	Industrial (FMEA)	Tabular FMEA data (HVS production)	Direct conversion from tables via set-theoretic schema	Hybrid: Cypher-first (LLM-gen), Vector fallback	GPT-4	Highly specialized to FMEA data structure. DFS chunking may miss distal context.
KG²RAG [11]	Multi-hop QA	HotpotQA (and "Shuffle" variant)	LLM-based triplet extraction from text chunks	Hybrid: Vector-first (seed chunks) + KG expansion	LLaMA3-8B	Multi-stage process adds latency; expansion can add noise that must be filtered.
Adaptive AI Tutor [5]	EdTech	Course Content (MIT Finance)	LLM-based extraction, expert-validated	KGR: Semantic search for nodes, then graph traversal	DeepSeek-V3	Efficacy tightly coupled to quality of the expert-validated KG.
KAQG [15]	EdTech (Question Gen.)	Educational Texts (ACT, etc.)	LLM extraction; PageRank for key concepts	RAG + Graph-based concept selection via PageRank	Not Specified (Multi-agent)	Complex multi-agent system; requires deep pedagogical (IRT) expertise.

vantages are a direct consequence of the structured reasoning and retrieval capabilities that knowledge graphs provide.

A. Analysis of performance improvements

The systems reviewed in this survey show a clear pattern: the integration of KGs into RAG pipelines leads to significant performance gains in a variety of domains and for different tasks. By employing structured knowledge, these systems are not limited to simple semantic retrieval but can perform more precise, context-aware, and logical operations. Table II summarizes the general performance increases.

TABLE II
PERFORMANCE COMPARISON OF KG-RAG FRAMEWORKS AGAINST BASELINES.

Framework	Baseline	Metric	Improvement
KGRA [8]	GPT-4	F1-Score (Keyword Ext.)	0.69 → 0.95
FMEA KG-RAG [14]	Std. RAG	Context Recall Accuracy	0.17 → 0.46
KRAGEN [16]	Std. RAG	(2-Hop QA) F1-Score	53.1% → 71.8%
KG ² RAG [11]	Semantic RAG	(Response) Mean Assess. Score	0.528 → 0.631
Adaptive AI Tutor [5]	Std. RAG		4.71 → 6.37

The data shows that fusing KGs with RAG is an effective strategy for improving factual accuracy, reasoning, or user performance. As an example, the KGRA [8] system's use of precise Cypher queries instead of vector search resulted in a 37.7% F1-score improvement for entity extraction compared to GPT-4. In a similar vein, KRAGEN's [16] application of GoTs for structured reasoning increased accuracy on complex biomedical queries by 18.7%. For education, the Adaptive AI Tutor [5] provided context-rich answers, leading to a 35.2% increase in average student test scores. Taken together, these results support the idea that the explicit, relational nature of

KGs offers the context and constraints needed to address the primary weaknesses of both LLMs and standard RAG.

Tables III and IV, along with Fig 6, provide a more detailed illustration of these gains using results from the KG²RAG framework [11]. The data in Table III indicates the final response quality; KG²RAG performed better than other methods on several datasets, particularly on the more complex HotpotQA-Full. This finding is reinforced by Table IV, which shows the high retrieval precision and recall of KG²RAG. This retrieval improvement stems directly from its KG-guided expansion and organization components. Lastly, Figure 6 shows the system's robustness, as KG²RAG's performance remains high across different values of k (retrieved chunks), whereas baseline models showed more sensitivity to this parameter.

TABLE III
F1-SCORE FOR RESPONSE QUALITY, COMPARING KG²RAG WITH BASELINES (FROM [11]).

Method	HotpotQA-Dist	HotpotQA-Full
LLM-only	0.237	0.237
Semantic RAG	0.617	0.528
+ Rerank	0.652	0.587
Hybrid RAG	0.653	0.551
LightRAG	0.293	0.261
GraphRAG	0.400	0.169
KG²RAG	0.663	0.631

IV. CONCLUSION AND FUTURE SCOPE

This survey has synthesized the rapidly emerging field of Knowledge Graph-enhanced RAG (KG-RAG). Our review confirms a clear trend: integrating structured KGs with LLM-based RAG systems yields substantial, measurable improvements over standard RAG baselines. We analyzed frameworks across diverse domains, from EdTech to cybersecurity, identifying a set of common strategies. These include

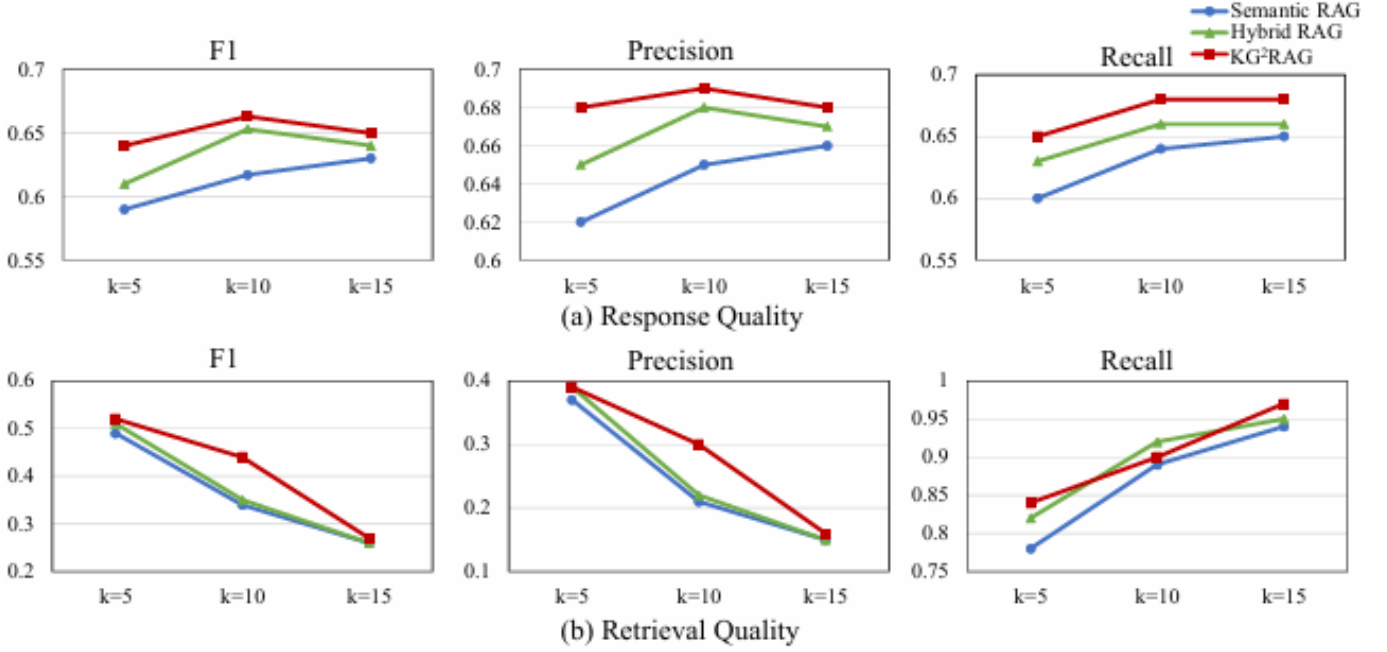


Fig. 6. Experimental results with varying top-k on HotpotQA in distractor setting. The charts show KG²RAG consistently outperforming Semantic RAG and Hybrid RAG on response quality (top row) and (bottom row) (from [11]).

TABLE IV
F1-SCORE FOR RETRIEVAL QUALITY, COMPARING KG²RAG WITH
BASELINES (FROM [11]).

Method	HotpotQA-Dist	HotpotQA-Full
Semantic RAG	0.343	0.300
+ RERANK	0.357	0.306
Hybrid RAG	0.354	0.302
LightRAG	0.234	0.132
GraphRAG	0.255	0.180
KG²RAG	0.436	0.310

hybrid retrieval (blending vector search and graph queries), precise graph-only retrieval (replacing vector search), and graph-guided context expansion. The quantitative evidence is compelling, with frameworks demonstrating significant gains in factual accuracy (e.g., an 18.7% rise in biomedical QA [16]), reasoning capability, and real-world user performance (a 35.2% increase in student scores [5]). The ‘Key Takeaways’ from our analysis highlight that the primary benefit of KGs is their ability to enforce explicit constraints, guide multi-hop reasoning, and provide coherent, interconnected context, thereby directly addressing the core RAG weaknesses of fragmented retrieval and shallow reasoning.

Despite this progress, our review identifies several open challenges and promising avenues for future research. The synthesis of frameworks in Table I reveals gaps in standardization, particularly in KG construction and the lack of a common LLM backbone for fair comparison. Key future directions include:

- **Dynamic and Automated KG Construction:** Current methods often rely on pre-existing or manually curated KGs. Future work should explore real-time, automated KG construction and updating (e.g., from streaming text data) to keep the RAG system’s knowledge current.
- **Adaptive Retrieval Strategies:** Most systems use a fixed retrieval strategy (e.g., hybrid, vector-first). Integrating Reinforcement Learning (RL) could allow the system to adaptively learn the optimal retrieval strategy (e.g., when to use a graph query vs. a vector search) based on query complexity and user feedback.
- **Multimodal KG-RAG:** This survey focused on text-based KGs. A significant frontier is the integration of multimodal data. Future systems could incorporate image, audio, and video information into the KG, enabling RAG to answer complex queries that span multiple data types (e.g., "Explain the process shown in the video using the concepts from the textbook").
- **Personalized and Explainable RAG:** In domains like EdTech, a ‘one-size-fits-all’ context is suboptimal. Future work could leverage user-specific KGs (e.g., modeling a student’s knowledge gaps) to provide personalized augmentation. Furthermore, the graph structure itself can be used to generate more intuitive, explainable answers, moving beyond ‘black-box’ LLM responses.

Addressing these challenges will be critical for developing the next generation of dependable, domain-aware, and personalized AI systems.

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